Buying New, Buying Second-Hand, or Renting?

A Life-Cycle and AHP-Based Decision Analysis of Formalwear for Students

Yang Yihua, Cao Xiaoyue, Yang Liu, Liu Ziqing, and Xie Tongling

Abstract

Students choosing whether to purchase or rent formalwear must balance carbon emissions, expenditure, and convenience. This study compares purchasing a new dress, purchasing a second-hand dress, and renting the same dress model for a National University of Singapore student attending six formal events over one year. Life cycle assessment (LCA) estimates allocated annual greenhouse gas emissions, while life cycle costing (LCC) quantifies direct consumer expenditure. These outcomes and a time-based convenience indicator are combined through analytic hierarchy process (AHP)-weighted composite scoring. Under baseline assumptions, annual emissions are 4.92 kg CO_2e for each purchase option and 11.38 kg CO_2e for rental. Annual expenditure is SGD 60.99, 72.00, and 108.00 for second-hand purchasing, new purchasing, and rental, respectively. Second-hand purchasing ranks first, reflecting its lower cost and identical modelled emissions and active time relative to new purchasing. Transport sensitivity analysis examines how collection-and-return travel affects the ranking. The results support second-hand purchasing in the baseline scenario and identify substantial reductions in attributed travel burdens as a condition for rental competitiveness.

Keywords: formalwear; life-cycle assessment; life-cycle costing; analytic hierarchy process; transport sensitivity

1 Introduction

Formalwear acquisition combines garment production, care, payment, and personal effort. Purchasing enables repeated use, while rental provides temporary access supported by collection, return, and cleaning services. Their relative performance depends on garment utilization and the operations required to serve each occasion. Clothing inventories illustrate the importance of production and use-phase assumptions (Sandin et al., 2019); research on formalwear rental further identifies use frequency and service arrangements as determinants of environmental performance (Monticelli & Costamagna, 2023).

This study compares three acquisition strategies for the Love, Bonito Eyka Multi-way Satin Column Dress, size S: buying new, buying second-hand, and renting for each event. A common annual scenario links carbon accounting and consumer expenditure to AHP-weighted evaluation. Transport sensitivity analysis then examines how collection-and-return arrangements influence the preferred option (Figure 1).

2 Methods

2.1 Scenario, evidence, and assumptions

The functional unit is the provision of six formalwear uses for one NUS student over one year. Each purchase option involves one acquisition and six wears; rental involves six separate seven-day bookings. The scenario permits repeated use of the same style and assumes suitable fit, dress-code compliance, hygiene, and availability. Figure 2 defines the common product system, its resource inputs, and its service and emission outputs. Manufacturing, distribution and retail, use and care, and eventual disposal follow the life-cycle grouping in formalwear LCA (Monticelli & Costamagna, 2023). Annual activities comprise cleaning and acquisition-related delivery or passenger travel. Packaging is reused, and the analysis focuses on the dress and its acquisition service.

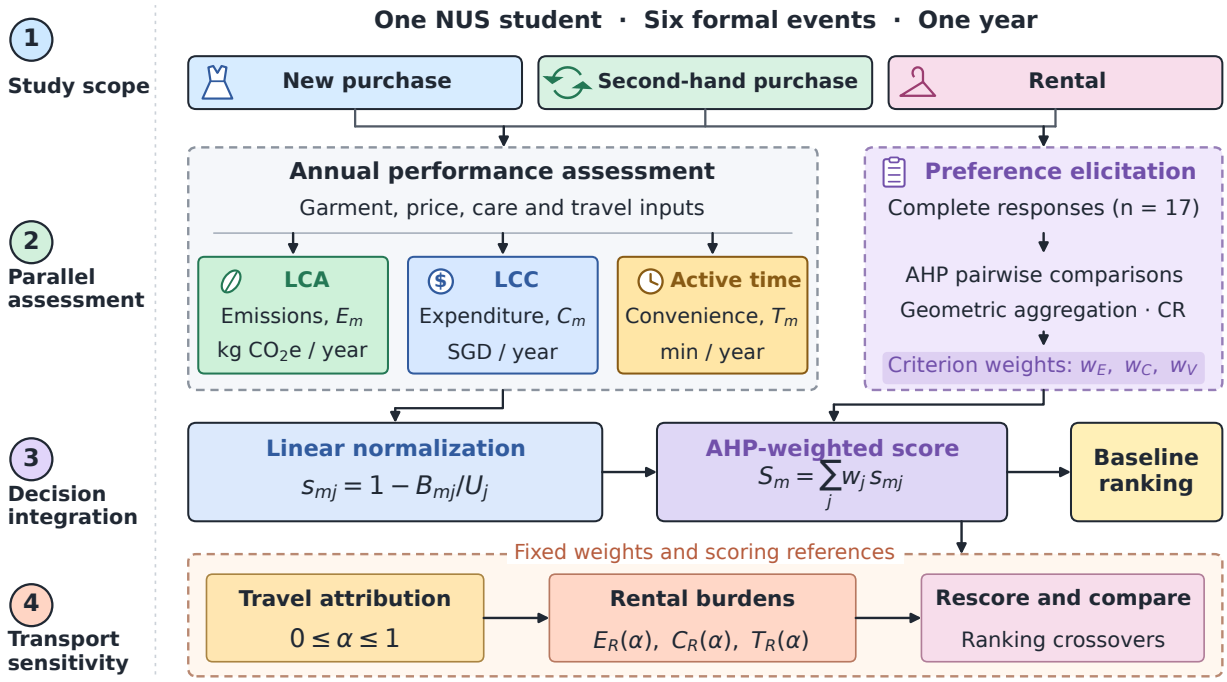


Figure 1: Research framework linking annual performance assessment and AHP weighting to baseline ranking and transport sensitivity analysis.

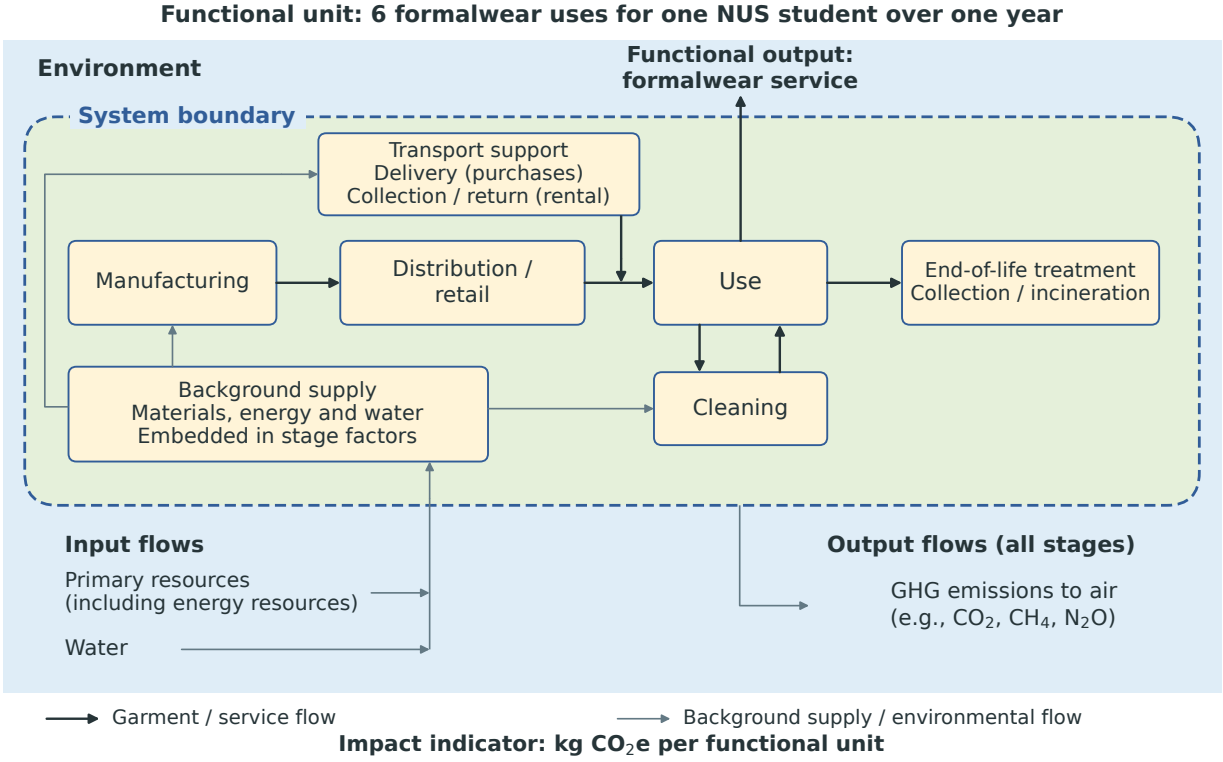


Figure 2: Product system and system boundary for the common functional unit. Resource and emission arrows represent qualitative flows; background supply is embedded in the stage factors. Climate-change impacts are quantified using aggregated literature factors and scenario-specific annual activities.

Advertised prices and terms were archived on 3 November 2026 from the manufacturer and two Singapore listings (forest_green66, n.d.; igotgr8deals, n.d.; Love, Bonito, n.d.). Garment-stage factors are rounded graphical extractions from Sandin et al. (2019), Figure 4.24. The cleaning proxy represents residential washing, drying, and ironing in an Italian study (Monticelli & Costamagna, 2023), Table 4; UK transport factors include direct and upstream energy emissions (Department for Energy Security and Net Zero, 2026). These proxies support a screening assessment. Table 1 summarizes the principal inputs. All channels share a 26-wear lifetime, and the second-hand garment's reported prior wear satisfies the remaining-capacity condition $1 + 6 \leq 26$.

Table 1: Principal inputs for the six-event annual scenario.

Input	Value and basis
Acquisition payments	New SGD 65; second-hand SGD 52; rental SGD 10 per seven-day booking (advertised)
Delivery and deposit	New delivery SGD 0; second-hand delivery SGD 1.99; refundable rental deposit SGD 99 (advertised)
Manufacturing / distribution / disposal	14.8 / 0.84 / 0.99 kg CO_2e per garment (literature proxies)
Cleaning	0.109 kg CO_2e per clean; purchases 7 cleans, rental 6 cleans (proxy and scenario)
Transport	Rail 0.0287 kg CO_2e/passenger-km; van 0.31844 kg CO_2e/vehicle-km (UK proxies)
Acquisition travel	Purchases: two 10-km delivery legs shared by 20 garment-equivalent items; rental: 24 legs of 10 km (scenario)
Care, fares, and active time	SGD 1 and 15 minutes per home clean; SGD 2 and 30 minutes per travel leg; 15 minutes per purchase order; 10 minutes per rental booking (scenario)

2.2 Annual emissions, expenditure, and time

Let $m \in \{N, S, R\}$ denote new, second-hand, and rental. With $n = 6$ annual wears, $L = 26$ lifetime wears, and per-garment manufacturing, distribution, and gross disposal factors M, D, F, annual emissions are

$$E_m = \frac{n}{L}(M + D + F) + k_m e_c + d_m e_v + p_m e_r, \tag{1}$$

where k_m denotes cleaning count, d_m allocated delivery vehicle-kilometres, p_m passenger-kilometres, and e_c, e_v, e_r the corresponding factors. The baseline uses $(k_N, k_S, k_R) = (7, 7, 6)$, $(d_N, d_S, d_R) = (1, 1, 0)$, and $(p_N, p_S, p_R) = (0, 0, 240)$. Garment-stage burdens are allocated equally across channels, with avoided-production and energy-recovery credits set to zero.

The delivery scenario assumes two 10-km legs shared equally by 20 garment-equivalent items, giving $d_N = d_S = (10 \times 2)/20 = 1$ allocated vehicle-km per purchase. Rental travel yields $p_R = 6 \times 4 \times 10 = 240$ passenger-km.

The LCC calculation measures direct annual consumer expenditure. Purchase payments enter in full, with resale receipts and residual values set to zero:

$$C_N = 65 + 7(1), \qquad C_S = 52 + 1.99 + 7(1), \qquad C_R = 6(10) + 24(2). \tag{2}$$

Rental care surcharges and additional platform charges are set to zero. The fully refundable SGD 99 deposit represents a temporary liquidity requirement. Carbon allocation and consumer expenditure thus reflect annual service use and annual cash payments, respectively.

Convenience is represented by active consumer time. Ordering and home care yield $T_N = T_S = 15+7(15) = 120$ minutes/year. Booking, handover, and dedicated travel yield $T_R = 6[10+4(30)] = 780$ minutes/year. These time inputs are scenario assumptions.

2.3 AHP weights and composite scoring

The original questionnaire contains 18 records, of which 17 provide complete comparisons of emissions (E), cost (C), and convenience (V). Q8, Q9, and Q10 supply the ratios E/C, E/V, and C/V.

Reciprocal matrices are aggregated geometrically with equal respondent weights (Forman & Peniwati, 1998). The group priorities are normalized row geometric means, which equal the principal eigenvector for a positive reciprocal 3×3 matrix:

$$\bar{a}_{ij} = \left(\prod_{r=1}^{17} a_{ij}^{(r)} \right)^{1/17}, \qquad g_i = \left(\prod_{j=1}^{3} \bar{a}_{ij} \right)^{1/3}, \qquad w_i = \frac{g_i}{\sum_h g_h}. \tag{3}$$

Individual and group consistency use $CI = (\lambda_{\max} - 3)/2$ and $CR = CI/RI$, with $RI = 0.52$ and criterion $CR \leq 0.10$ (Saaty, 1994). All 17 complete responses contribute to the group weights, providing exploratory preference estimates.

Annual burdens $B_{mj} \in \{E_m, C_m, T_m\}$ are converted to linear scores and combined using these weights:

$$s_{mj} = 1 - \frac{B_{mj}}{U_j}, \qquad S_m = w_E s_{mE} + w_C s_{mC} + w_V s_{mV}, \tag{4}$$

with annual reference values $(U_E, U_C, U_V) = (24\ \text{kg}\,\text{CO}_2\text{e}, 360\ \text{SGD}, 1080\ \text{minutes})$. Higher scores indicate preferred outcomes. The reference values set the scales of the linear functions, which retain their form across the burden domain.

2.4 Transport sensitivity analysis

Let $\alpha \in [0, 1]$ denote the share of baseline rental collection-and-return travel attributed to rental. Dedicated journeys correspond to $\alpha = 1$; fully shared journeys with zero incremental travel burden correspond to $\alpha = 0$. Travel emissions, fares, and active time change proportionally, while rental payments, cleaning, and handling remain fixed:

$$\begin{aligned} E_R(\alpha) &= 4.4916923 + 6.888\alpha, \\ C_R(\alpha) &= 60 + 48\alpha, \\ T_R(\alpha) &= 60 + 720\alpha. \end{aligned} \tag{5}$$

Ownership outcomes, criterion weights, and scoring references remain fixed. Rental–ownership crossover shares are obtained by equating their composite scores.

3 Results

3.1 Annual burdens and baseline ranking

Both purchase options yield 4.92 kg CO_2e/year, compared with 11.38 kg CO_2e/year for rental (Figure 3). The common garment-stage allocation contributes 3.84 kg CO_2e/year. Rental travel contributes a further 6.89 kg CO_2e/year, accounting for approximately 61% of its total emissions. Annual expenditure is SGD 60.99 for second-hand, SGD 72.00 for new, and SGD 108.00 for rental (Table 2). Rental payments total SGD 60, with travel adding SGD 48 and 720 active minutes.

The AHP weights are 0.2410 for emissions, 0.4295 for cost, and 0.3295 for convenience. Eleven of 17 individual matrices satisfy $CR \leq 0.10$, and the group matrix has $CR = 0.005885$. With these weights, second-hand purchasing scores 0.8412, followed by new purchasing at 0.8281 and rental at 0.5189 (Table 3).

Second-hand purchasing saves SGD 11.01 relative to new purchasing while sharing the same modelled emissions and active time. New purchasing has lower burdens than rental on all three criteria. These relationships establish the baseline ordering for positive criterion weights.

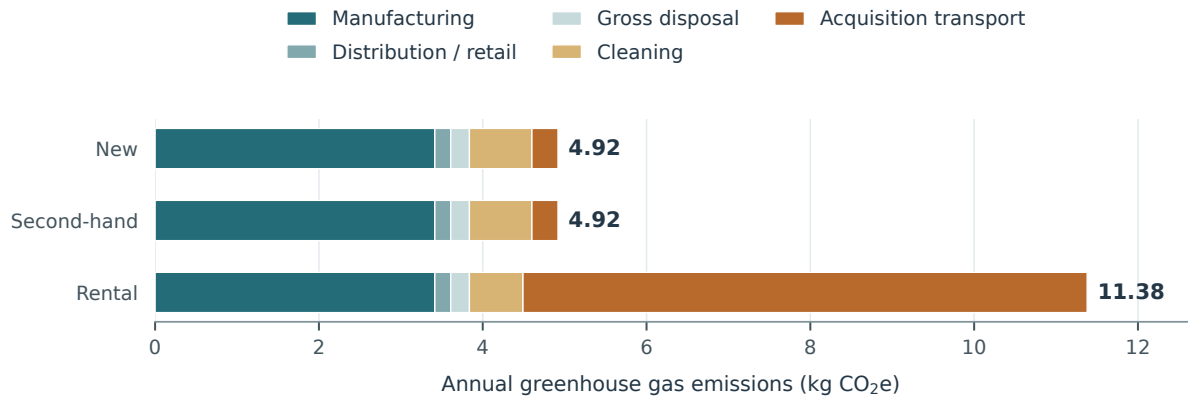


Figure 3: Annual greenhouse gas emissions by life-cycle stage. All channels use the common 26-wear garment allocation.

Table 2: Annual consumer expenditure by component (SGD).

Option	Garment	Delivery	Rent	Care	Travel	Total
New	65.00	0.00	0.00	7.00	0.00	72.00
Second-hand	52.00	1.99	0.00	7.00	0.00	60.99
Rental	0.00	0.00	60.00	0.00	48.00	108.00

Table 3: Baseline annual outcomes, criterion scores, and composite ranking.

Option	kg CO_2e/year	SGD/year	Minutes	s_E	s_C	s_V	S (rank)
Second-hand	4.92	60.99	120	0.7950	0.8306	0.8889	0.8412 (1)
New	4.92	72.00	120	0.7950	0.8000	0.8889	0.8281 (2)
Rental	11.38	108.00	780	0.5258	0.7000	0.2778	0.5189 (3)

Note. Minutes denote assumed active consumer time. Calculations retain full precision.

3.2 Transport and ranking changes

Substitution into Eq. 4 gives

$$S_R(\alpha) = 0.865006510 - 0.346091054\alpha. \tag{6}$$

Rental equals second-hand at $\alpha = 0.068704$ and new at $\alpha = 0.106662$ (Figure 4). Accordingly, rental ranks first below approximately 6.87% of baseline attributed travel burdens, second between approximately 6.87% and 10.67%, and third above approximately 10.67%. The intersecting alternatives share equal scores at each boundary.

4 Discussion

The second-hand advantage reflects a lower acquisition price under common garment allocation, care, and delivery assumptions. Equal emissions for the purchase options follow from this allocation convention. Rental's repeated collection and return generate substantial travel costs and time, explaining its baseline position. Its first-place crossover near 7% identifies extensive travel sharing as a condition for competitiveness within the proportional scenario. Collection integrated with existing journeys therefore warrants attention in rental service design.

The estimates depend on the selected garment lifetime, advertised prices, proxy inventories, and assumed active time. The Italian care and UK transport factors provide screening estimates for the Singapore setting. Product-specific inventories, local routes, actual transaction costs, and channel-specific lifetimes would strengthen empirical precision. The original survey supports a

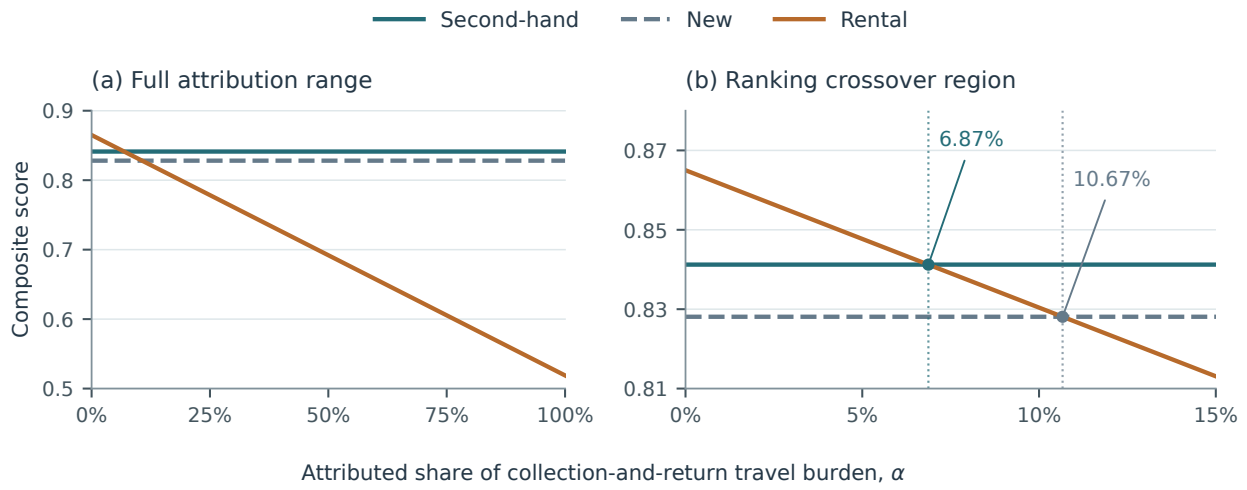


Figure 4: Composite scores as the attributed share of rental collection-and-return travel changes. Criterion weights and ownership outcomes remain fixed.

small-sample assessment of criterion preferences; individual and collective CR values characterize their respective judgment matrices. Transport thresholds also depend on the scoring references and the proportional relationship among travel emissions, fares, and time. These conditions define the interpretation of the modelled ranking.

5 Conclusion

For one student attending six formal events annually, second-hand purchasing achieves the highest AHP-weighted score, followed by new purchasing and rental. Its advantage arises from lower expenditure under equal modelled purchase-related emissions and time. Rental becomes the highest-scoring option when its attributed collection-and-return travel burdens fall below approximately 6.87% of the dedicated-travel baseline. The findings support second-hand acquisition under the specified assumptions and link rental competitiveness to efficient collection-and-return arrangements.

Data and reproducibility

The repository retains the original questionnaire, parameter and source registers, respondent-level CR values, and full-precision results. The scripts `code/run.py` and `report/generate_assets.py` regenerate numerical outputs and manuscript assets; execution instructions accompany the repository. The archived evidence records source locations and input hashes.

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AI use statement. AI assistance supported English revision, LaTeX preparation, figure generation, and computational and document checks.

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Bibliographic note. Citation years follow the proceedings volume for Saaty (1994) and the journal volume for Monticelli and Costamagna (2023). Their online publication dates are June 1, 2009 and 2022, respectively. The source register provides the archived market and inventory evidence.